Knowledge organiser · extraction steps

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Extraction means separating material from cells. A membrane is a boundary around a cell or an organelle.

Mashing breaks up soft fruit tissue. Detergent helps disrupt cell and nuclear membranes. This releases a mixture of cell contents.

Salt assists extraction and precipitation. Filtering removes larger pieces of tissue. The liquid that passes through is the filtrate.

Cold ethanol helps DNA precipitate. A precipitate is material that comes out of a solution. The trained adult handles and layers the ethanol.

Use only the approved method. No tasting or direct pupil contact with chemicals. A supplied-observation route is an equal choice.

Keep each role clear. Say what each step does. Use one step at a time.

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Version _____ 1

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Knowledge organiser · observation and limits

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

An observation records what was seen. Say where the material was and what it looked like.

White strands may be consistent with a DNA-containing precipitate. This uses method knowledge as well as sight. It is an interpretation.

A crude extract is a mixture that has not been purified. Visible material may include other substances. Appearance alone does not establish purity.

A water comparison helps test whether strands appeared in every tube. It does not identify every substance in the fruit sample.

Do not use visible amount as a true measure of DNA yield. Keep fruit mass, volumes, mixing and timing fair. An appropriate DNA measurement is needed for true yield.

Fruit-cell chromosomes contain DNA and genes. This extraction cannot show an individual gene, base sequence or double helix to the naked eye.

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Word help · extraction and evidence

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Extraction: separating material from cells. Extract: remove from a mixture.

Membrane: a boundary around a cell or an organelle. Detergent: helps disrupt membranes and release cell contents.

Filter: separates larger debris from liquid. Filtrate: the liquid that passes through the filter.

Precipitate: material that comes out of a solution. Crude extract: a mixture that has not been purified.

Observation: what you saw. Interpretation: what the evidence may mean, using what you know.

Purity: whether other substances are present. Yield: how much of a substance is obtained. Visible amount alone does not measure true DNA yield.

DNA carries genetic information. A gene is a section of DNA. Chromosomes contain long DNA molecules. The extraction does not show a single gene or its base sequence.

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Arrival · Supported · four clues

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

1. Release: What does detergent help to disrupt?

Help: Read or point first. Cell and nuclear boundaries.

2. Filter: What does filtering remove?

Help: Read or point first. Large tissue pieces or individual genes?

3. Observe: Is “white strands appeared” an observation or a purity test?

Help: Read or point first. It describes what was seen.

4. Safety: Who handles and layers the cold ethanol in this route?

Help: Read or point first. Use the adult-directed method.

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Arrival · Independent · four clues

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

1. Release: What does detergent help to disrupt?

Help: Cell and nuclear boundaries.

2. Filter: What does filtering remove?

Help: Large tissue pieces or individual genes?

3. Observe: Is “white strands appeared” an observation or a purity test?

Help: It describes what was seen.

4. Safety: Who handles and layers the cold ethanol in this route?

Help: Use the adult-directed method.

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Arrival · Aspire · four clues

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

1. Release: What does detergent help to disrupt?

Help: Cell and nuclear boundaries.

2. Filter: What does filtering remove?

Help: Large tissue pieces or individual genes?

3. Observe: Is “white strands appeared” an observation or a purity test?

Help: It describes what was seen.

4. Safety: Who handles and layers the cold ethanol in this route? Give a reason.

Help: Use the adult-directed method.

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Starter · choose a first idea

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

White, stringy material appears. Which statement is most careful?

1. I can see one gene.

2. A crude DNA-containing extract may have formed.

3. This proves the sample is pure DNA.

Think first. Give one choice and a reason before checking the evidence. Keep the first idea so you can review it later.

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Model 1 · Break up the tissue

Today I can

Explain release, filtering and precipitation steps.

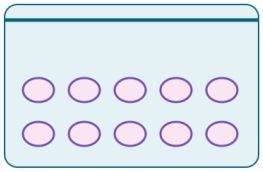
Record an observation without claiming purity.

State one practical limit or useful comparison.

Gently mash soft fruit in a sealed bag with the centre-approved extraction solution. Use gentle pressure. An adult prepares the fruit and controls materials. No tasting.

This breaks up tissue and exposes cells. The picture is a simplified model, not a photo of your own result.

Start with soft tissue. Keep the bag sealed. Use gentle pressure. This is an adult-led method. You may direct the adult.



Mash: break up tissue

Sealed bag • soft

fruit • gentle

pressure

Adult preparation • no tasting

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Model 2 · Release the contents

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Detergent helps disrupt membranes. Salt assists the extraction and precipitation process.

The membranes around the cell and nucleus are boundaries. Releasing cell contents gives a mixture, not a purified DNA sample.

A membrane is a boundary. It keeps some contents inside. Breaking that boundary lets contents out. The contents still form a mixture.



Detergent: disrupt membranes

Membranes around cell and nucleus

Released cell contents: a mixture

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Model 3 · Remove large debris

Today I can

Explain release, filtering and precipitation steps.

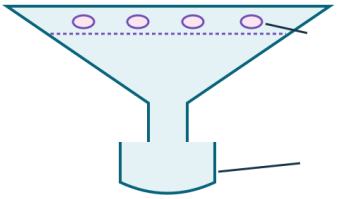
Record an observation without claiming purity.

State one practical limit or useful comparison.

Filter the mixture. The liquid contains dissolved and suspended material released from the fruit.

Filtering removes large tissue debris. It does not remove every dissolved substance. The liquid that passes through is the filtrate.

A filter traps large bits. Liquid goes through. Some small or dissolved material may go through too. Clear does not mean pure.



Filter: remove large debris

Large debris

Filtrate

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Model 4 · Observe the interface

Today I can

Explain release, filtering and precipitation steps.

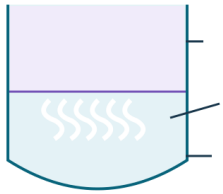
Record an observation without claiming purity.

State one practical limit or useful comparison.

An adult carefully layers cold ethanol. White, stringy material may appear; it is a crude DNA-containing extract.

Observe the boundary between liquids. Appearance alone does not establish DNA purity. Other material may remain. No flames, sparks or heat.

Look where the two liquids meet. Say what you see. Keep the claim small. White strands can be a mix of substances.



Ethanol: observe a crude precipitate

Cold ethanol

White strands

Fruit filtrate

Adult handles

ethanol. No

flames or heat.

Appearance alone does not establish DNA purity.

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Sort · six extraction and evidence cards

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Choose a card, then a group. Explain one placement using its clue. Take turns as chooser and reason-questioner.

The groups are Release contents / Separate material / Observe and evaluate.

1. Mash soft fruit
2. Add approved detergent–salt solution
3. Filter the mixture
4. Adult layers cold ethanol
5. Record white stringy material
6. Name one contamination limit

Point, use keys, move a card or direct an adult. Swap roles after three cards. Each learner keeps one own reason.

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Sort · card clues and a choice of help

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

1. Mash soft fruit: Break up tissue and expose cells.

2. Add approved detergent–salt solution: Disrupt membranes and assist extraction.

3. Filter the mixture: Remove larger tissue debris.

4. Adult layers cold ethanol: Help DNA precipitate from solution.

5. Record white stringy material: State the appearance without claiming purity.

6. Name one contamination limit: Crude extract may contain other material.

Would a small change help? Ask the lesson teacher for clearer labels, fewer cards, reading help, an adult-directed model or more thinking time.

The teacher will explain what can change, try it and check its effect. Ask privately or pass. No extra form is needed.

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Worked example · a clear liquid

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

After filtering, a liquid looks clear. Does that prove it contains only DNA?

1. The observation is appearance: the liquid looks clear.
2. Filtering removed large debris, not every dissolved substance.
3. Other molecules may remain even when they are not visible.
4. Therefore a clear filtrate is not evidence of pure DNA.

Use what the method can show. A clear look alone cannot identify all the substances present.

Look at the liquid first. Say what you saw. Then say what that can show. The look alone cannot tell us what every substance is.

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Practical or supplied evidence · safe access

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Use only the teacher-approved method. A trained adult handles all liquids and layers cold ethanol. You may direct an adult or use the supplied observations instead.

Teacher or technician approves the method, concentrations, quantities, eye protection and disposal. Use adult-prepared soft fruit, sealed bags and a tray.

Check allergies privately. No public disclosure is needed. No tasting, human samples or mouth pipetting. No direct pupil contact with chemicals.

Keep ethanol away from flames, sparks and heat. Follow local chemical risk controls. Wash hands. Staff clear spills and dispose of materials as approved.

Observation, adult-directed demonstration and supplied-data routes are equal choices. Do not treat the supplied records as your own practical observations.

You can use the paper route. You do not need to touch the kit. Direct the adult if you wish. The science task stays the same.

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Supplied observations · keep the evidence route clear

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

These observations are fictional teaching records. They are not results from your classroom. Use them if a practical or demonstration is not suitable.

Fruit extract: white stringy material at the liquid boundary after layering ethanol.

Water comparison: no visible strands after layering ethanol.

Limit: no chemical purity test was performed.

The observation is consistent with a DNA-containing precipitate from the fruit extract. The water comparison helps show that strands did not simply appear in every tube.

It does not identify all substances or prove purity. Record an actual demonstration only if you observed it. Keep supplied and observed evidence distinct.

A record tells us what was seen. A claim says what it may mean. Keep them apart. The sample has a clear limit.

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Evidence record · observation, meaning and limit

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Label the evidence route: classroom observation, adult demonstration or supplied fictional record.

Record the ordered method. Say what mashing, detergent, salt, filtering and adult-controlled ethanol each do.

Write “I observed…” for what you saw. For supplied evidence, use “The supplied record states…”. Describe colour, location and texture.

Then give a cautious interpretation: “This is consistent with a crude DNA-containing extract.” Do not call it pure DNA.

Name one limit and a useful comparison or improvement. A water comparison does not identify every component.

Connect the material to fruit-cell DNA, chromosomes and genes. The extraction does not reveal individual genes or a base sequence. Keep the evidence and its limit together.

Use three short parts. First say what you saw. Next say what it may mean. Then say what it cannot show.

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DNA, genes and chromosomes · recap and connection

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Use this recap if it helps. Do not assume a previous lesson was completed. Describe complementary base pairing. State the chromosome–DNA–gene relationship.

DNA carries genetic information. A gene is a section of DNA. A chromosome contains a long DNA molecule. DNA bases pair in a model; this practical cannot show those pairs.

The extracted DNA came from fruit-cell chromosomes and contains many genes. It does not reveal an individual gene or the base sequence.

Explain what filtration separates. Why is an observation different from identifying every component?

Keep a precise method, an observation and one evaluation point. A diagram is simplified and not to scale. It is not a photograph of your own result.

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Reference cards · observation, inference or limit

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Use within the evidence discussion if needed. Classify a statement, then explain the clue. This is not another practical.

1. White strands: white stringy material appeared at the liquid boundary.

2. DNA-containing material: the method and expected chemistry support that the precipitate contains DNA.

3. Purity unknown: the white material may contain substances other than DNA.

Groups: Direct observation / Scientific inference / Method limitation. Say what was seen, what is inferred and what cannot be concluded.

Take turns to choose, justify and check. You may direct a partner to move a card. Cutting is optional.

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Independent task · Supported

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Choose one route. Use the whole shared 14 minutes for your task, feedback and one improvement. Do not complete all three routes.

1. Explain: Link mashing, detergent, filtering and ethanol to their roles.

Help: Use the step cards: mash / detergent / filter / cold ethanol. Match each step to its stated purpose.

2. Record: Write an observation and a cautious interpretation using your evidence route.

Help: Choose actual demonstration or supplied observations. Complete: I observed ___; this may indicate ___.

3. Evaluate: Give one limitation and one useful comparison or improvement.

Help: Complete: the appearance supports a crude extract, but does not prove ___. One useful extra check would be ___.

Label classroom or supplied evidence. Adult handles all materials. Use paper, cards, point or exact scribing. Keep your own explanation and its limit.

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Independent task · Independent

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Choose one route. Use the whole shared 14 minutes for your task, feedback and one improvement. Do not complete all three routes.

1. Explain: Link mashing, detergent, filtering and ethanol to their roles.

Help: Use release, remove debris and precipitate.

2. Record: Write an observation and a cautious interpretation using your evidence route.

Help: Start “I observed…” or “The supplied record states…”.

3. Evaluate: Give one limitation and one useful comparison or improvement.

Help: Purity, equal fruit mass, same volumes or a water comparison.

Label classroom or supplied evidence. Adult handles all materials. Use paper, cards, point or exact scribing. Keep your own explanation and its limit.

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Independent task · Aspire

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Choose one route. Use the whole shared 14 minutes for your task, feedback and one improvement. Do not complete all three routes.

1. Explain: Link mashing, detergent, filtering and ethanol to their roles.

Help: Use release, remove debris and precipitate.

2. Record: Write an observation and a cautious interpretation using your evidence route.

Help: Start “I observed…” or “The supplied record states…”.

3. Fair comparison: Two groups get different amounts of visible material. Why is this not automatically a fair comparison of DNA yield?

Help: Think fruit mass, volumes, mixing, timing and purity.

Label classroom or supplied evidence. Adult handles all materials. Use paper, cards, point or exact scribing. Keep your own explanation and its limit.

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Review · real audience and one next move

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Show one useful science explanation to the adult. Keep your words and evidence route exact. The adult names back the science they heard or saw.

Check the method role, the observation, the interpretation and the limit. Change one overclaim or clarify one missing link.

Choose a named next move: secure / mixed / misconception / access / method or evidence. Say “Next I will ___.” Try the change on your existing work.

Keep what you said and what changed after the adult listened. No extra participation form is needed. Review happens within the 14-minute independent block.

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Exit · Supported · two checks

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Answer both small checks. Use your own explanation and the model or evidence.

1. Role: Why is detergent used in fruit-DNA extraction?

Help: Use: my answer is ___ because ___. Point to a model or card first. Think membranes and released contents.

2. Limit: Why should the visible material not be called pure DNA?

Help: Use: the evidence shows ___, but cannot yet show ___. Consider what else may be present.

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Exit · Independent · two checks

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Answer both small checks. Use your own explanation and the model or evidence.

1. Role: Why is detergent used in fruit-DNA extraction?

Help: Think membranes and released contents.

2. Limit: Why should the visible material not be called pure DNA?

Help: Consider what else may be present.

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Exit · Aspire · two checks

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Answer both small checks. Use your own explanation and the model or evidence.

1. Role: Why is detergent used in fruit-DNA extraction?

Help: Think membranes and released contents.

2. Fair comparison: Two groups get different amounts of visible material. Why is this not automatically a fair comparison of DNA yield?

Help: Think fruit mass, volumes, mixing, timing and purity.

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